Andrew Ma

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Education

The University of Texas at Austin — Austin, TX
Bachelor of Science in Computer Science | GPA: 3.3/4.0 — *Aug. 2025 – May 2029*
- **Activities:** Freshman Research Initiative, Autonomous Driving Group, Pi Kappa Alpha Fraternity
- **Coursework:** Data Structures & Algorithms, Discrete Math, Calculus III, Probability, Linear Algebra

Experience

Software Contractor — Jul. 2026 – Present
BizPro Software (LLC) — Clients: Allied IT Systems — *Beaumont, TX*
- Built React, TypeScript, and .NET asset-management portal managing **5,000+** endpoints across **60+** client sites from one console
- Engineered Redis-queued webhook integration to PSA/RMM ticketing, processing **1,200** events per minute at **sub-90ms** p99 latency
- Cut API error rate **85%** (2.1% to 0.3%) via SQL Server query indexing and **88%** backend test coverage

Software Engineering Intern — May 2026 – Jul. 2026
Port of Port Arthur — *Port Arthur, TX*
- Fused **6** AIS telemetry streams into a unified PostGIS spatial state at **sub-second** refresh across **40+** tracked vessels
- Designed a conflict-safe schema with `btree_gist` exclusion constraints preventing overlapping berth reservations across **100+** bookings
- Created spatial data infrastructure and automated scheduling systems, transforming vessel streams into predictive maritime analytics.

Research Analyst, Autonomous Driving Group — Jan. 2026 – Present
The University of Texas at Austin — Advised by Dr. Nathan Tsoi — *Austin, TX*
- Architected a multi-agent V2V framework broadcasting shared costmaps across racing fleets at sub-100ms latency
- Optimized multi-sensor perception via Joint Entropy Reduction to lower localization uncertainty in fused pose estimates
- Trained neural driving policies on **50K+ frames** with IL/RL, deploying inference on Jetson Orin in live track trials

Associate Researcher — Oct. 2025 – May 2026
SALT Lab, UT Austin — Advised by Dr. David Harwath and PhD Student Yuan Tseng — *Austin, TX*
- Co-authored a paper published to the **IEEE Spoken Language Technology Workshop (SLT 2026)**.
- Built a deepfake detection pipeline reaching **85% accuracy** across **6 benchmarks** (audio + text, **10K+ samples**)
- Engineered BERT + few-shot transformer ensembles with adversarial refinement to improve robustness under distribution shift

Research Intern — NLP, ML & Software Engineering — Feb. 2025 – Aug. 2025
Lamar University — Advised by Dr. Rakibul Islam and Dr. Stephan Andrei — *Beaumont, TX*
- Co-led an LLM/Transformer study classifying **1,000+ software requirements**; **published in IEEE Access (2025)**
- Benchmarked Zero-Shot vs. Few-Shot prompting across 5 tasks, reaching **F1 of 94%** on multi-class NFR/security classification
- Fine-tuned ResNet-50/DenseNet on **5,000+ CT/MRI scans** (**10GB+ DICOM**) to **94.2%** accuracy, with a Grad-CAM layer surfacing model attention for clinicians

Projects

libsbf-rs Contributions — Saronic Technologies (Open Source) | *Rust, binrw, proptest* — 2026
- Added a multi-message streaming property test covering the previously-untested core parse path, verified under **4,096** cases
- Refactored fixtures onto a shared framing helper, removing duplicated CRC/length logic across the test suite
- Retyped raw `u8` fields as existing enums across **4** message types via `binrw` maps, with zero wire-format change

WakeUp — Maritime Vessel Spoofing Detection | *Python, PyTorch, PyTorch Geometric, scikit-learn, PostGIS* — 2026
- Built a data generator simulating **5** types of spoofing attacks on ship-tracking signals to benchmark detection models
- Engineered a Transformer model flagging subtle spoofing attacks, raising detection accuracy from **64%** to **97%** over baselines
- Designed an evaluation harness scoring detection precision and false-alarm rates across **4** competing models

GhostHive.net — Decentralized AI Agent Platform | *Node.js, PostgreSQL, Knex.js, Socket.io, Solana* — 2026
- Built a marketplace of **100+** autonomous agents that discover, negotiate, and transact tasks with state sync over Socket.io
- Designed a **Solana-backed digital currency** as the settlement layer, letting agents pay each other on-chain for completed work
- Engineered a Node.js + PostgreSQL backend with Knex.js coordinating agent actions, escrow, and $2,000+ in transactions

Honors & Awards

USACO Gold (Top **4.13%**) | **NMSI STEM Star Scholarship** (2 of 2,000) | **FPS International Scholar** (4 of 30,000)
TMSCA Science 1st Place State | **AP Scholar with Distinction** | **Home to Texas Scholar (UT Austin)** (40/600)

Technical Skills

Languages: Python, Java, C, C++, C#, JavaScript, TypeScript, SQL, Bash, Rust, Go
ML / AI: PyTorch, TensorFlow, Transformers, LLMs, RAG, OpenCV, scikit-learn, LangChain, NLP
Robotics & Systems: ROS2, Nav2, Gazebo, MPPI, SLAM, Jetson Orin, IL/RL, Linux
Backend & Data: Node.js, FastAPI, Flask, PostgreSQL, PostGIS, Redis, GraphQL, REST API, WebSockets
Cloud & DevOps: AWS (EC2, S3, Lambda, RDS), Docker, CI/CD (GitHub Actions), Nginx